Do Highly Engaged YouTube Videos Have More Extreme Comments?

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Introduction, Background & Significance

With 2.5 billion monthly active users, YouTube is one of the largest platforms for online interaction. Its comment sections are a major site for social interaction between viewers and creators. Yet, comments are often treated only as an engagement metric rather than a reflection of the tone and quality of public discourse. Prior research suggests that emotionally intense or extreme content tends to generate higher engagement, raising concerns that platform incentives may amplify divisive or hostile discourse. However, it remains unclear whether higher engagement is actually associated with more extreme comment sections, particularly at scale and across different content types. Our project addresses this knowledge gap by constructing a dataset that links video metadata (views, likes, comments) to comment-level sentiment and toxicity scores for video-level analysis.

In YouTube comment sections, viewers interpret and discuss content with other viewers and the creator using critique, humor, and conflict. By dissecting the comment section, you can observe how norms evolve in language, belief, and culture. From a social computing perspective, this makes comments not just an engagement signal, but a clear reflection of community sentiment at the moment. By quantifying sentiment across videos, our study treats comment sections as a social system where tone and structure can be measured and compared.

This matters because YouTube is an engagement-driven platform, where visibility and reach are shaped by clicks, likes, and comments. These engagement signals are not neutral as they can shape expectations before participants click a video, influencing what is “normal”, what feels worth saying, and how strongly users are supposed to react. If higher engagement reliably corresponds to more extreme or hostile environments, that creates a potential feedback loop: content that triggers strong reactions attracts more interaction, which can increase visibility and invite further viewership from the algorithm. Understanding whether these engagement cues relate to discourse sentiment is crucial for explaining how the platform design can indirectly influence public conversation.

These findings raise broader implications for platform governance. If certain metadata patterns are associated with heightened toxicity, platforms can deploy targeted interventions to prevent this behavior, such as adjusting how engagement metrics are displayed, how the algorithm ranks top comments, and investing more in comment moderation. Beyond YouTube, we hope to explore broader questions in social computing about how attention-optimized systems shape online hostility, particularly in domains such as news and politics.

Related Work

We have based our work on five works relevant to our research question, ranging from research experiments that provide methodological foundations to papers that serve as analytical frameworks. Through field experiments, Beknazar-Yuzbashev et al. (2023) ascertain that toxic content increases engagement across Facebook, Twitter, and YouTube (implying a trade-off between engagement and curbing toxicity). Our study investigates the inverse question, instead measuring whether extreme comments are more engaged; however, our results indicate no relationship between the two. Mili et al. (2024) reach a conclusion similar to that of Beknazar et al., claiming that YouTube's algorithm amplified extreme posts, resulting in higher engagement. Our study advances this in a meaningful way by investigating the connection between engagement and extremeness using a larger/more diverse dataset. Importantly, our null correlation challenges the premise of the engagement-amplification hypothesis they put forth (at least for YouTube). Lorentz and Singh (2021) investigate how well overall sentiments expressed in YouTube comment sections predict the number of likes for the corresponding video, finding that comment sentiment is not entirely informative about engagement (low correlation). Notably, the authors point out a sampling bias resulting from their dataset's abnormally high average like proportion, which they note inflated the correlations for predictions. Our study, in contrast, observed a link between comment extremeness and engagement and found a null result that aligns with Lorentz and Singh's line of thinking, despite correcting for the bias by using a dataset with high variation in comment engagement (number of likes). Amin et al. (2025) and Dehkoda et al. (2023) both present machine learning models and techniques that enabled them to examine the relation between user engagement and sentiment extremity. Our study directly extends these previous works by developing a scalable, construct-driven methodology (capable of handling >600,000 data points) for analyzing extremity in online discourse and using more precise sentiment analysis frameworks (VADER and Detoxify).

The sources we chose provided a solid foundation for our research, but we addressed some of their limitations in this study. Primarily, most existing works addressing this topic (such as Lorentz and Singh) quantify engagement by total likes, which introduces a bias towards more popular content (a comment on a popular video will naturally have more likes than one on a less popular one). Furthermore, there appear to be gaps in generalizability and sampling: Amin et al. (2025) trained their models on gaming-related video data, while Dehkhoda et al. (2023) employed a very small dataset (only about 3700 comments). We remedied this by using a much larger dataset spanning multiple genres.

Overall, our project is a partial replication of existing studies on sentiment-engagement metrics in YouTube's US dataset, adopting a combination of analytical methods under varying constraints to generate novel data. The main departure we have taken from existing literature is to conduct a more comprehensive, cross-categorical analysis spanning many YouTube spaces, including channels of varying sizes, target audiences, and genres (as opposed to a more specific examination such as Amin et al.). Additionally, while Hoiles et al. (2017) focus on how metadata predicts video popularity, our work extends this line of inquiry by linking video-level engagement and metadata cues to the tone and extremeness of the resulting comment discourse. Specifically, we have investigated the average extremeness of comments across different video categories, namely: sports, film & animation, music, etc. Another significant deviation we have made is our multi-dimensional analysis of users' engagement with YouTube comments. Our analysis centers around an "engagement score", a single summary metric we have defined to represent the overall engagement a video receives. Conceptually, it is the average of the normalized counts of views, likes, and overall comments for a video, providing a well-rounded picture of a video's engagement. Altogether, our related works are characterized by small datasets, narrow analysis scopes, and simplified illustrations of engagement; our study addresses these gaps through an expansive and diverse dataset, a cross-categorical analysis, and a multi-dimensional assessment of engagement.

Objective, Goals & Outcomes

The goal of this project is to determine whether YouTube videos that attract a lot of attention tend to receive more extreme or intense comments than those with less attention. To do this, we have set the following goals:

- Measure how extreme comments are on a YouTube video by looking at whether they are either strongly positive or strongly negative. This will be completed utilizing NLP tools.
- Measure video engagement by considering counts of views, likes, and comments to create a composite engagement score.
- Link comments to the videos they belong to so we can see how extreme a video's comments are relative to its popularity, enabling easier analysis of engagement and discussion patterns.
- Look for patterns across different types of videos (using YouTube API category IDs mapped to categories such as Music, Sports, and News & Politics) to determine if some channels and video topics have more extreme comments than others.

By the end of this project, we expected to produce insights into the link between engagement and comment intensity that would directly answer our research question. Through this work, we will know whether videos that get more views, likes, and comments also tend to have more extreme or intense comments. With these insights, we hoped to produce visualizations that show patterns cleanly, such as charts and graphs illustrating how engagement and comment extremeness relate. Specifically, we will create scatter plots to show the relationship between overall engagement scores and comment extremeness and box plots to illustrate the distribution of comment extremeness by video category. We will have generated a clean, merged dataset, with all video metadata and comment data processed and combined into a single new dataset. The remainder of the report presents key findings, with an emphasis on patterns in comment extremeness, differences across categories, and how our results extend prior research. For example, one way our project extends prior research is by reversing the direction of analysis of Lorentz and Sigh (2021), who tested whether comment sentiment predicts the number of likes on a video. Moreover, we will extend the work of Mili et al. (2024) by expanding the scope of engagement and extremeness research to the comment level, as the original paper examined the original posts themselves.

Description of Work Accomplished: Data

The dataset used for this research was “YouTube-Comments-Datasets” by Beytullah Soylev, available on Kaggle. This data was last updated 2 years ago. The data was collected by downloading the dataset from Kaggle and programmatically retrieving additional metadata using the YouTube Data API. This dataset contains numerous JSON and CSV files, but our group primarily focused on uscomments.csv, which contained over 691,350 comments with 2,006 unique video IDs. On average, each video contained approximately 305 comments, though the distribution varied widely, with some videos having only one comment and others up to 800. The dataset includes videos across multiple YouTube categories, with a higher concentration in Music (199) and Entertainment (188), and lower representation in categories such as Travel & Events (7) and Autos & Vehicles (11), indicating an uneven distribution across content types. We programmatically retrieved additional metadata using a Python script that made YouTube API calls to match each video ID to its corresponding video title, channel title, category ID, publish time, views, likes, and comment count. This step was necessary because the Kaggle dataset lacked complete engagement metadata, so the YouTube API was used to supplement the missing information required to construct engagement measures. This dataset was appropriate for addressing the research question because comments constitute direct user-generated behavior on YouTube, a highly engaged

platform. Comments were important for this project because they provide textual data that can be analyzed with sentiment and toxicity models, which are necessary to construct the extremeness metric. Unlike views or likes, comments capture the content and intensity of user expression, making them appropriate for this analysis. YouTube was selected because it is a highly interactive platform where engagement and discussion are closely linked.

The data was also publicly available, which made it practical to use. After collection, preprocessing steps were performed to ensure data quality and usability, including dropping missing comments, converting text to strings, and handling bad lines. One major issue with this dataset was its large size. This resulted in slow processing during analysis, so batching was used when running Detoxify, and an initial sample was processed for testing before the full pipeline was run. Another challenge was that not all video IDs returned valid results from the YouTube API, leading to missing metadata. To address this, we filtered out incomplete entries to ensure consistency in the final dataset.

Description of Work Accomplished: Approach

To accomplish our goals, we used a computational, data-driven approach that transformed YouTube comments and video metadata into measurable features for large-scale comparison. Because the data consisted of unstructured text, our approach focused on converting comments into quantitative features for comparison across videos. This directly aligned with our objective of testing whether videos with higher engagement also had more extreme comment sections. A feature-based representation was used to treat human expression as measurable signals. Each YouTube comment was converted into structured numerical features, specifically sentiment and toxicity scores. Since “extremeness” is not directly observed, it was constructed using two main inputs: sentiment and toxicity. Sentiment was measured with Valence Aware Dictionary and sEntiment Reasoner (VADER), which produced a compound score representing emotional polarity and intensity. We used the absolute value of the compound sentiment score so that strongly positive and strongly negative comments were both treated as emotionally intense. Toxicity was measured with Detoxify, which produced a toxicity score representing harmful or aggressive language. We then averaged sentiment strength and toxicity to create an extremity score for each comment, weighting them equally to capture emotional intensity and harmfulness as complementary aspects of extreme expression, without privileging one dimension over the other. A multi-level analysis was then completed. At the micro level, each individual comment was assigned a sentiment, toxicity, and extremeness score. We then grouped comments by video_id and calculated each

video’s average comment extremeness. At the macro level, each video became a unit of analysis with an engagement score and average extremeness score. Because views, likes, and comment counts were highly skewed, we applied log transformations and normalization, followed by min-max normalization to each feature, and then computed the engagement score as the average of three normalized values (views, likes, and comments). This links micro-level behavior to macro-level outcomes.

This approach was selected because the research question focuses on patterns across videos, requiring a method capable of analyzing relationships at scale. The computational approach was appropriate because it enables pattern detection across large datasets. This approach also allowed for consistency and objectivity because the same model was applied to all comments. This reduces human bias in coding and ensures comparability. One major strength of this approach is that it provides a consistent way to measure extremeness across a large number of comments and videos. Because each video's comments were scored using the same models and scoring logic, the resulting scores were comparable across videos. Another advantage is that the approach integrates multiple signals into a single measure. This method does not rely solely on positive/negative (sentiment) or toxic language to determine an overall measure of a user's expression; rather, it combines both measures to capture all possible aspects of a user's expression, making it more robust than approaches that rely on a single metric. Finally, the pipeline is applicable to various platforms and/or data sets, because the same process can be used to assess if these phenomena are observed in different settings.

One major limitation is construct validity. In our approach, extremeness is constructed rather than directly observed. Sentiment does not always equal intensity, and toxicity does not always equal extremeness. Because of this, our extremeness metric should be interpreted as a proxy for extreme expression rather than a complete measure of all forms of extremeness. Another limitation is the loss of context in text analysis. Each comment is analyzed in isolation, so the approach ignores conversation threads, replies, and timing, potentially missing the social context of these interactions. Lastly, there is a platform and sampling bias. The Kaggle dataset only includes YouTube comments, and the commenters are not representative of all viewers, so certain voices are overrepresented. This means the results may not generalize beyond YouTube or active commenters. Our findings may not generalize to platforms like X, where repost-driven algorithms may amplify emotionally charged content differently than YouTube’s video-centered discussions.

Description of Work Accomplished: Results

We analyzed 691,350 comments across 1,968 unique YouTube videos to test whether higher-engagement videos are associated with more extreme comment environments. Our primary finding was that they are not related, or at least not in the direction predicted by prior work on engagement-driven amplification. Across the full sample, the Pearson correlation between the composite engagement score and average comment extremeness is r = -0.06 (R^2 = 0.00), indicating no linear relationship at the pooled video level. However, this pooled null result masks substantial variation across content categories, and the category itself proves to be a far stronger predictor of comment extremeness than any engagement signal we measured.

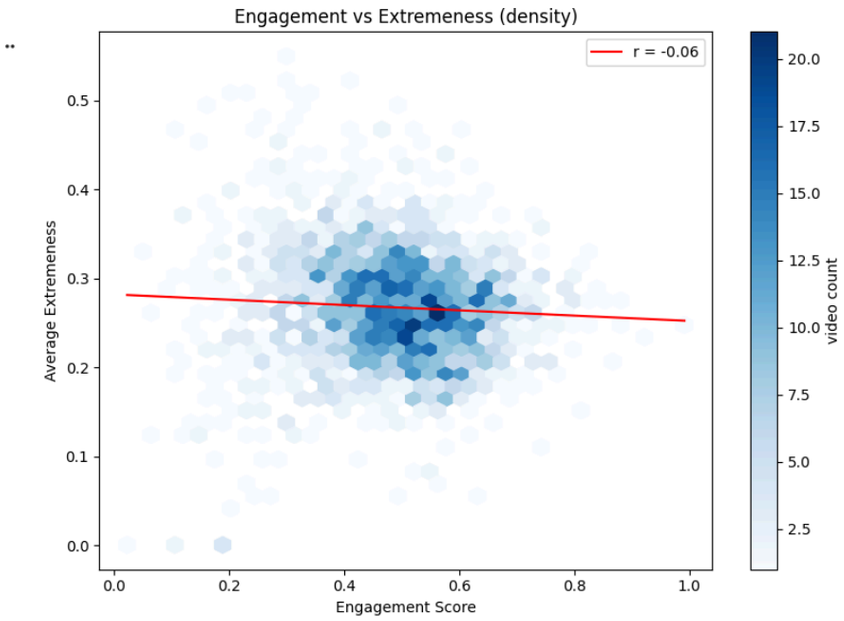


Figure 1: A hexbin density plot that shows the relationship between engagement score and average comment extremeness.

As seen in Figure 1, the mass of videos sits in a tight cluster at moderate engagement (0.4 to 0.6) and moderate extremeness (0.22 to 0.32), with no evidence of non-linear structure, bimodality, or a cluster of highly engaged, highly extreme videos. The fitted regression line is nearly flat with a slight negative slope. We also note that the cloud is somewhat asymmetrical, with low-engagement videos showing greater dispersion in extremeness than high-engagement videos. High-engagement content also appears to have a soft floor on extremeness around 0.20, suggesting that videos do not become widely engaging while remaining emotionally flat, though this asymmetry does not translate into a meaningful linear relationship.

To check whether the pooled null result might be obscured by how engagement is constructed, we examined the relationships among individual engagement components and their associations with the

average extremeness. Figure 2 presents the correlation heatmap. The three engagement variables (log views, log likes, and log comments) are highly intercorrelated (ranging from 0.86 to 0.92), and each correlates with the composite engagement score above 0.96. This indicates that all four measures capture a single underlying construct of video popularity and that no individual component carries information substantially different from that of the composite. Each component's correlation with average extremeness ranges from -0.10 to 0.00, with none meaningfully different from zero. Thus, the components individually correlate with extremeness no more than the composite does, ruling out the possibility that aggregating views, likes, and comments into a single score masked the real relationship.

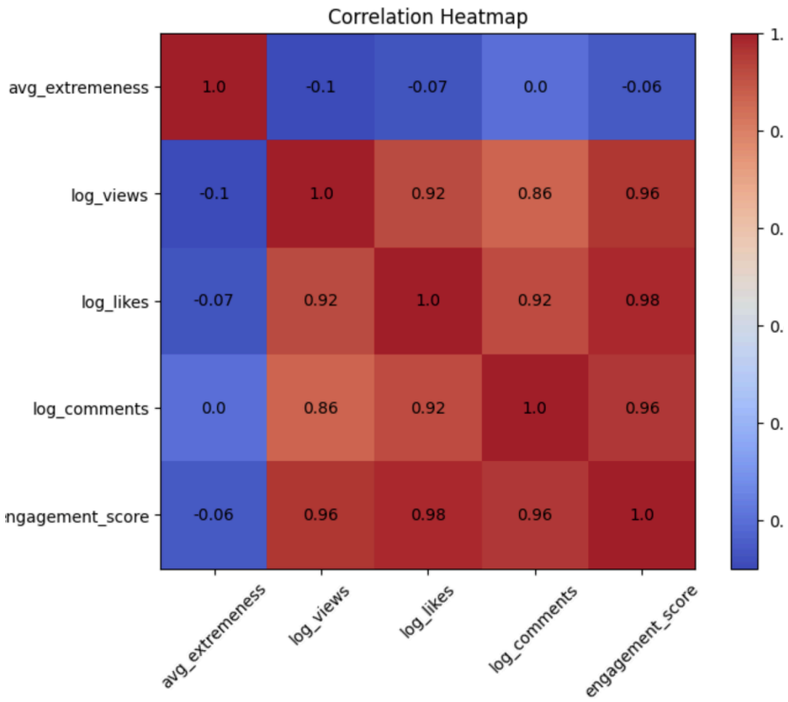


Figure 2: Correlation heatmap among the three engagement components (log views, log likes, log comments), the composite engagement score, and average comment extremeness across 1,968 videos.

The three engagement components are highly intercorrelated (r = 0.86 to 0.92), and each correlates with the composite above 0.96, indicating a single underlying construct. None of the engagement variables correlates with average extremeness above |0.10|.

To understand whether the pooled null result obscures category-specific dynamics, we examined the engagement-extremeness relationship within each category. Figure 3 below presents per-category scatter plots for six representative categories. Looking at the data, we can assess heterogeneity across categories. Music and News & Politics show subtle negative trends, while Entertainment and Comedy show clouds with no apparent relationship. This per-category variation motivates closer data analysis of

both between-category differences in extremeness and within-category correlations between engagement and extremeness, which we will explore further.

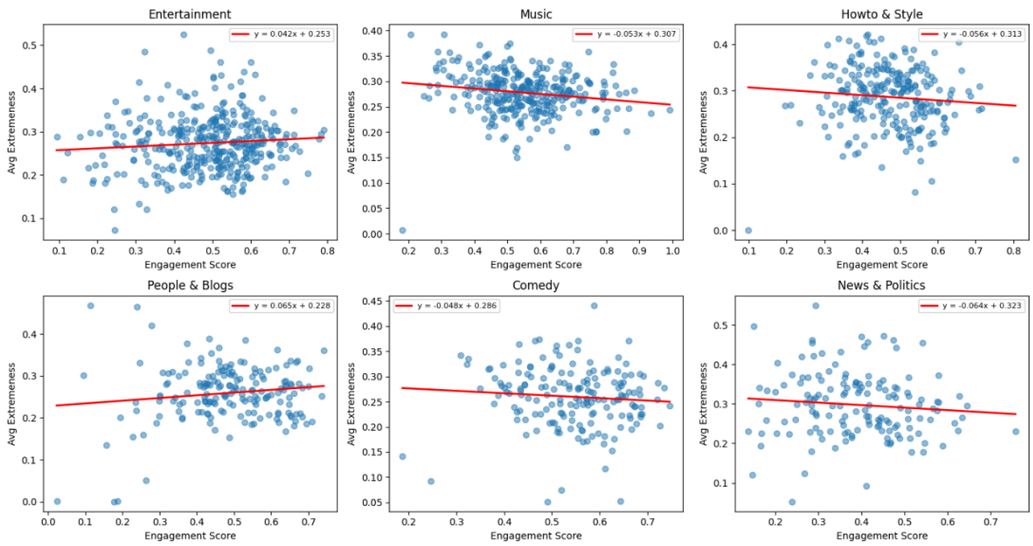


Figure 3: Per-category scatter plots of engagement score against average comment extremeness

Content category emerged as the strongest predictor of average comment extremeness in the dataset. Figure 4 presents the mean average extremeness by category, sorted in ascending order. Mean extremeness ranges from 0.220 in Gaming to 0.298 in News & Politics, a spread of 0.078 between categories. This variation within the categories exceeds the variation in extremeness associated with engagement, both in the pooled sample and within every category we analyzed. News & Politics, Howto & Style, Music, and Sports occupy the upper end of this distribution, while Gaming, Science & Technology, Education, and Autos & Vehicles cluster at the lower end. This pattern is intuitive, since topic selection appears to set the baseline tone of comment sections more than any video-level engagement signal.

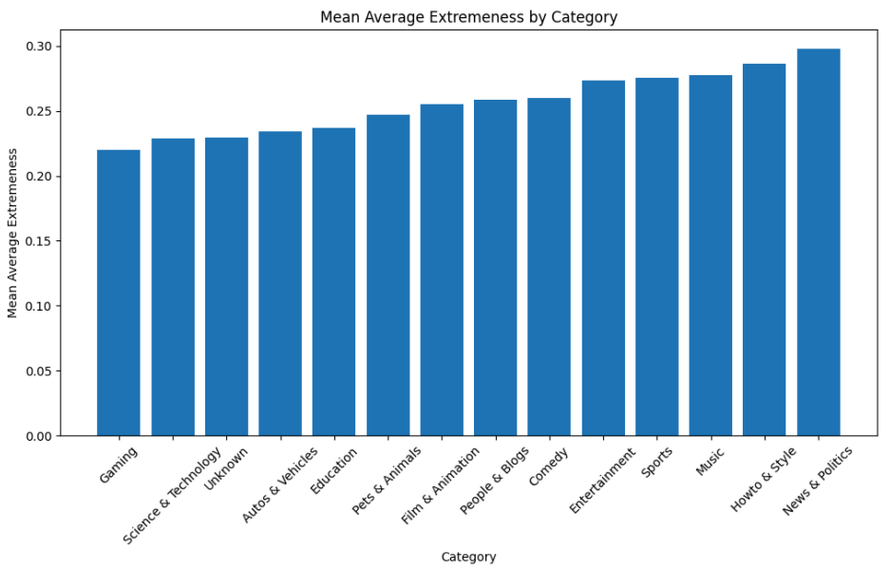


Figure 4: Mean average comment extremeness by content category, sorted in ascending order

Per-Category Engagement–Extremeness Correlations
(* p < 0.05, ** p < 0.01)

Category	n	r	p	Mean Extremeness	Sig.
Sports	113	-0.281	0.003	0.276	**
Film & Animation	88	-0.216	0.043	0.255	*
Music	314	-0.169	0.003	0.278	**
Education	87	-0.151	0.162	0.237	
Science & Technology	144	-0.145	0.083	0.229	
News & Politics	150	-0.095	0.249	0.298	
Howto & Style	248	-0.088	0.167	0.286	
Comedy	174	-0.082	0.283	0.260	
Entertainment	370	+0.084	0.106	0.274	
People & Blogs	178	+0.133	0.077	0.259	
Pets & Animals	34	+0.158	0.371	0.247	

Figure 5: Within-category Pearson correlations between engagement score and average comment extremeness

We also explore the within-category Pearson correlations between engagement score and average extremeness for all categories with at least 30 videos. The numerical results reveal a consistent asymmetry, with significant negative relationships. Three categories show statistically significant negative correlations. Sports ($r = -0.281$, $p = 0.003$, $n = 113$) demonstrated the strongest effect, followed by Film & Animation and Music. In each of these categories, more popular videos are associated with calmer rather than more extreme comment environments. Education and Science & Technology show similar negative correlations that approach but do not reach significance thresholds. Most importantly, no category shows a statistically significant positive correlation. The two closest categories, Pets & Animals and People & Blogs, are not significant and have small sample sizes. The most notable result concerns News & Politics, which exhibited the highest mean extremeness. We expected this category to be affected by engagement-driven amplification, as predicted by Milli et al. and Beknazar-Yuzhashev et al, but we found no such effect (2024; 2023). The within-category correlation was $r = -0.095$, which is statistically indistinguishable from zero. Thus, highly engaged political video comment sections are not measurably more extreme than less engaged ones, suggesting that the elevated extremeness of political content is a property of the topic itself rather than the function of how widely the content circulates.

Discussion of Outcomes, Implications, and Conclusion

Our findings provide a nuanced perspective on the relationship between engagement and discourse on YouTube. Contrary to our initial hypothesis and much of the prior literature, we find no

meaningful relationship between video engagement levels and the extremeness of their comments. This suggests that highly viewed or liked videos are not systematically associated with more emotionally intense comment environments. The story becomes more interesting when we examine relationships within categories. Where engagement correlates with extremeness, the relationship runs in the opposite direction from what prior research on amplification accounts reported: more popular videos in Sports, Film & Animation, and Music draw less extreme commentary, and no category in our sample shows a statistically positive correlation. This challenges a common assumption in social computing that engagement signals inherently amplify extreme or polarizing content. Prior work has shown that emotionally charged posts can drive interaction, but our findings suggest this may not translate cleanly to YouTube comments. We've found that even if extreme content attracts attention, that does not necessarily mean the overall surrounding discussion becomes more extreme. Our analysis highlights that the content category plays a much more substantial role in shaping comment extremeness than engagement metrics do. For example, videos in categories such as News & Politics consistently exhibited higher extremeness scores than those in categories such as Gaming and Science & Technology. This suggests that the topic or context of a video establishes the norms for discourse about it, rather than the extent to which the video is viewed or interacted with. One question our findings raise is why, within categories like Sports, Music, and Film & Animation, more popular videos tend to attract less extreme commentary rather than more. Several explanations are plausible. Highly engaged videos in these categories may attract a broader, more casual audience, whose comments dilute the more intense reactions of niche viewers, lowering the per-video average. Alternatively, creators of widely viewed content may also invest more in moderation.

Overall, these findings carry implications for how we think about platform incentives and online discourse. Regarding how platform design shapes the tone of comment sections, our results suggest that it does so primarily through topic selection, meaning what kinds of videos get made and recommended, rather than through a uniform amplification of extreme content as engagement rises. This complicates a straightforward feedback loop reading of engagement and hostile discourse on YouTube. For platform governance, interventions that focus on engagement signals, such as adjusting how popularity cues are displayed or how comments are ranked, may be less effective than category-aware moderation strategies that account for the distinct discourse norms of different content domains. Future work should examine whether these patterns hold at the comment-thread level, where reply dynamics and conversational drift may produce different results than the video-level analysis we have conducted.

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